

# Personal AI Infrastructure Design

## Objective

Build a **Personal AI Infrastructure** that:

- Learns from my personal context.
  - Can run mostly on my **MacBook M1 Pro (16GB RAM)**.
  - Uses cloud AI only when necessary.
  - Is accessible from my phone.
  - Supports MCPs (Model Context Protocol).
  - Automatically chooses the best AI model for each task.
  - Optimizes for cost while providing excellent AI capabilities.
  - Stays within **₹2,000–₹3,000/month**.
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## Hardware

Current Machine:

- **MacBook Pro M1 Pro**
- **16 GB Unified Memory**
- SSD Storage

This machine is powerful enough to:

- Run local LLMs (7B–14B comfortably)
- Run embeddings
- Host vector databases
- Run automation
- Host MCP servers
- Perform image analysis
- Act as the AI gateway

It is **not** suitable for replacing frontier models like Claude or GPT-5.5 for complex reasoning.

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# Primary Use Cases

## 1. Apple Music AI

### Goals

- Learn my music taste.
- Automatically create playlists.
- Discover new music.
- Update playlists weekly.
- Download music automatically (where supported).
- Organize my library.

### Workflow

```
Prompt
  ↓
AI
  ↓
Apple Music MCP
  ↓
MusicKit API
  ↓
Playlist Creation
  ↓
Automatic Updates
```

## 2. Photo Organization

### Goals

- Analyze 10GB+ of photos locally.
- Rank images by quality.
- Detect duplicates.
- Create meaningful albums.
- Generate reels automatically.

### Local Processing

```
Photos Folder
  ↓
CLIP Embeddings
  ↓
```

Image Tagging  
↓  
Aesthetic Scoring  
↓  
SQLite Metadata

Each image stores metadata such as:

- People
- Location
- Landscape
- Beach
- Food
- Pets
- Portrait
- Night
- Colorfulness
- Blur score
- Aesthetic score
- Duplicate similarity

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## Album Creation

Example:

Prompt:  
  
"Show me the Goa trip."  
  
↓  
  
Vector Search  
  
↓  
  
Relevant Images  
  
↓  
  
Create Album

## Reel Generation

Prompt

↓

Top Photos

↓

Video Clips

↓

Music Recommendation

↓

FFmpeg

↓

Final Reel

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## 3. Coding Assistant

### Goals

- Startup development
- Architecture planning
- Code generation
- Refactoring
- Reviews
- Documentation

### Model Selection

Small Tasks

→ Local Model

Medium Tasks

→ GPT

Large Architecture

→ Claude

Huge Refactors

→ Claude Code

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Recommended Local Models

- Qwen Coder
  - DeepSeek Coder
  - Gemma
  - Llama
- 

## 4. Personal Assistant

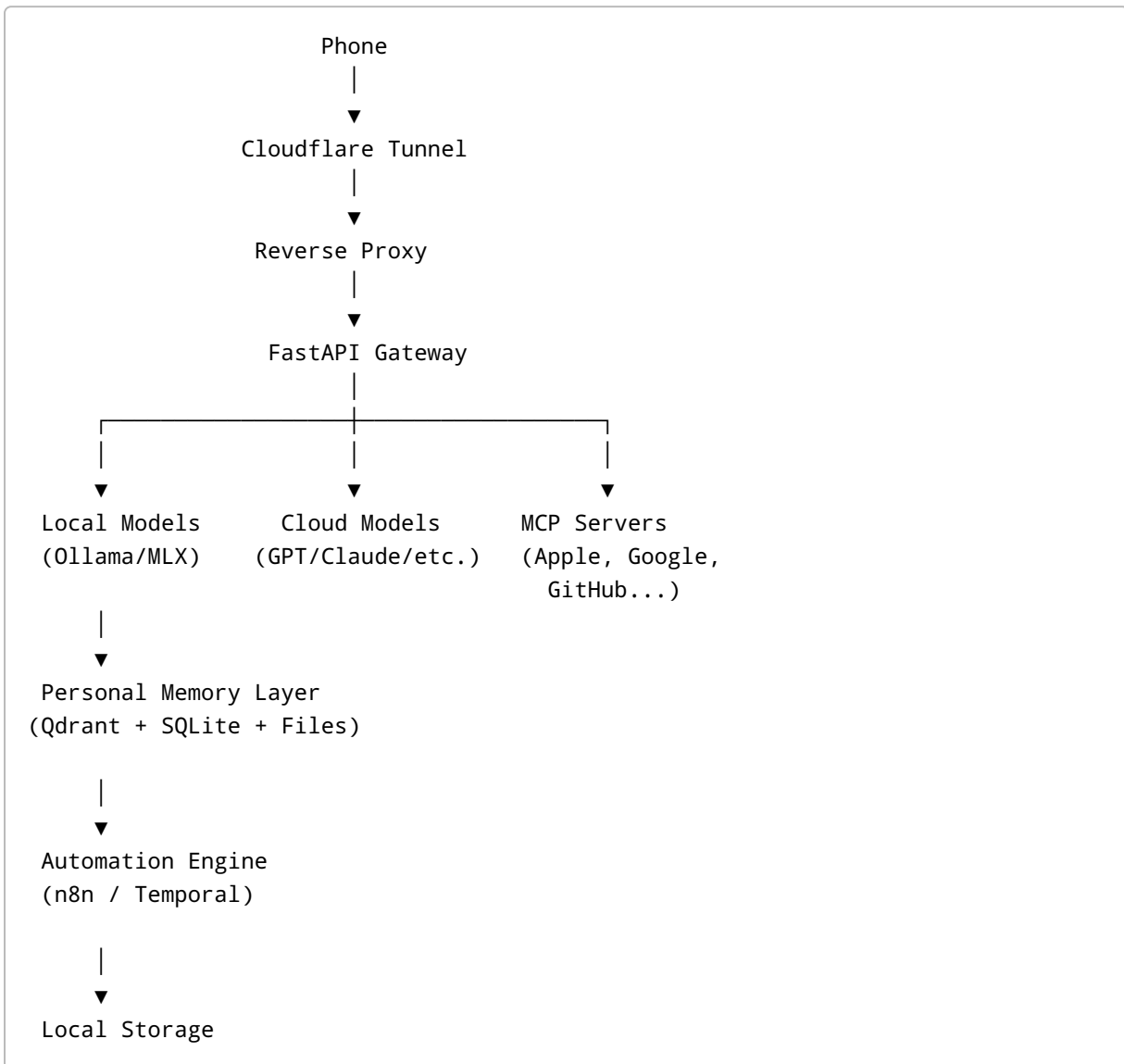
Capabilities

- Notes
  - Calendar
  - Email
  - PDFs
  - OCR
  - Web browsing
  - Research
  - Summaries
  - Personal memory
- 

## 5. Miscellaneous

- Automation
  - GitHub
  - PDFs
  - Meeting notes
  - Travel planning
  - Shopping research
  - Expense analysis
-

# High-Level Architecture



## Hosting Options

### Option A — Fully Local

#### Components

- Ollama
- MLX
- SQLite

- Qdrant
- FastAPI
- n8n

#### Pros

- Unlimited usage
- Maximum privacy
- Zero cloud costs

#### Cons

- Lower reasoning capability
- Phone access requires exposing Mac

#### Rating

★★★★☆

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## Option B — Hybrid (Recommended)

#### Components

- Local models
- Cloud models
- Local memory
- Cloudflare Tunnel
- MCPs

#### Pros

- Best intelligence
- Low cost
- Private data
- Excellent coding
- Great scalability

#### Rating

★★★★★

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## Option C — Fully Cloud

### Pros

- Always available
- Reliable
- Easy scaling

### Cons

- Expensive
- Everything uploaded
- Privacy concerns

### Rating

★★★★☆☆

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## Local AI Stack

### LLM Runtime

- Ollama
  - MLX
- 

### Models

#### General

- Gemma
- Llama
- Mistral

#### Coding

- Qwen Coder
- DeepSeek Coder

#### Vision

- CLIP
- Florence
- SigLIP



## Embeddings

- BGE
- Nomic Embed

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# Personal Memory

SQLite

+

Qdrant

+

Embeddings

+

Markdown Notes

+

JSON Metadata

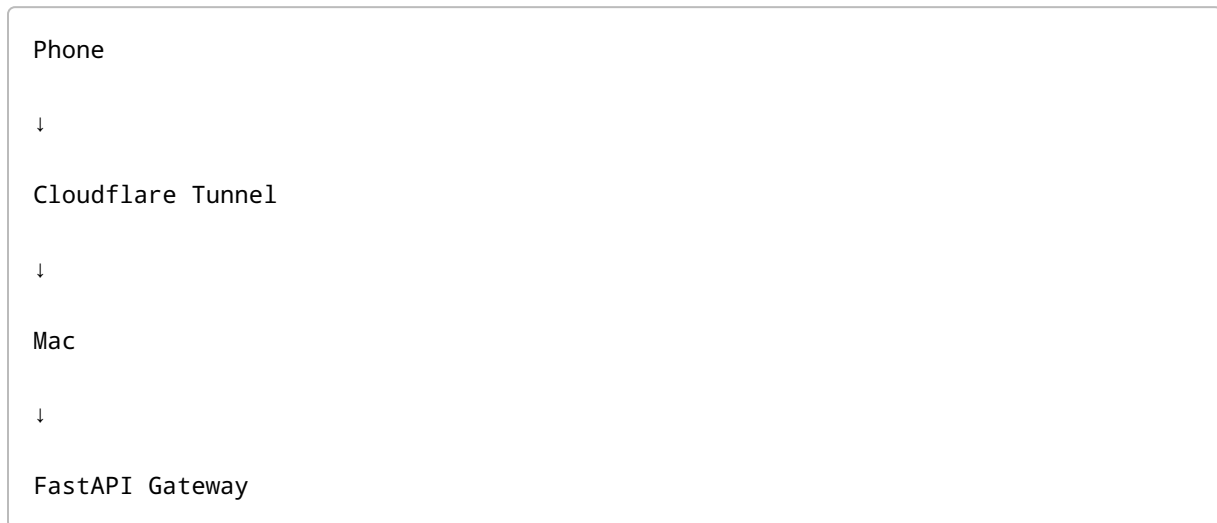
+

Git Repository

## Stores

- Conversations
- Projects
- Preferences
- Music taste
- Photos
- Coding context
- Documents

# Phone Access

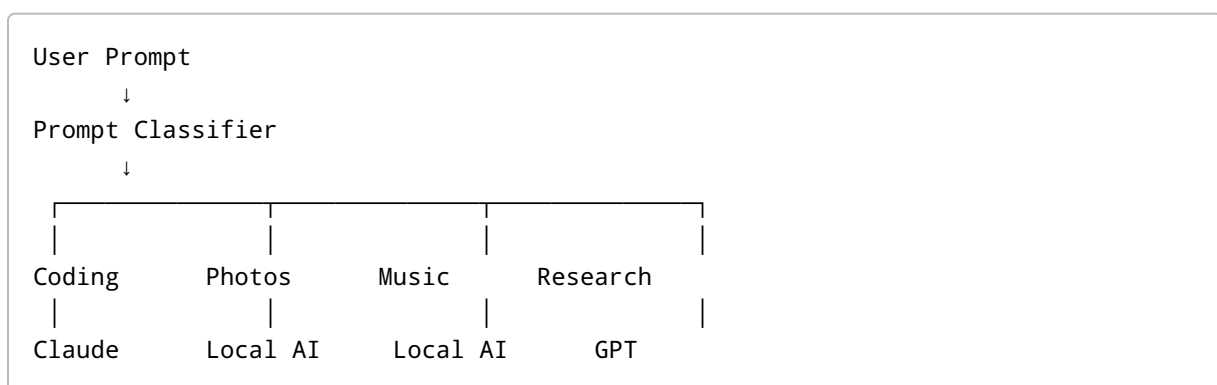


## Benefits

- No port forwarding
- Secure
- Works anywhere

# Model Router

The system should automatically choose the appropriate AI.



## Example

Simple question

→ Local Model

Simple coding

→ Local Qwen

Large coding

→ Claude

Deep research

→ GPT

Image analysis

→ Local Vision Model

Music organization

→ Local Model

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## Web Research Stack

Recommended

- Playwright
- Firecrawl
- Crawl4AI
- Browser Use

Workflow

Prompt

↓

AI Browses Web

↓

Reads Sources

↓

Summarizes

↓

Stores Results

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## Image Ranking Pipeline

Photos

↓

CLIP

↓

Aesthetic Predictor

↓

Blur Detection

↓

Face Detection

↓

Duplicate Detection

↓

Ranking

↓

Album Suggestions

↓

Reel Suggestions

## Output

- Best 100 Photos
  - Best Portraits
  - Best Landscapes
  - Best Food
  - Best Travel
  - Duplicate Detection
  - Delete Suggestions
- 

# Recommended Technology Stack

## Backend

- FastAPI

## Automation

- n8n

## Local Models

- Ollama
- MLX

## Memory

- SQLite
- Qdrant

## Embeddings

- BGE
- Nomic Embed

## Vision

- CLIP
- Florence

## Reverse Proxy

- Cloudflare Tunnel

## Authentication

- OAuth
- JWT

## Database

- SQLite

## Object Storage

- Local SSD

## APIs

- Apple MusicKit
- Google Photos
- GitHub
- Gmail
- Calendar

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## Estimated Monthly Cost

| Component                | Monthly Cost |
|--------------------------|--------------|
| Electricity              | ₹100-300     |
| Ollama                   | ₹0           |
| MLX                      | ₹0           |
| SQLite                   | ₹0           |
| Qdrant                   | ₹0           |
| n8n                      | ₹0           |
| Cloudflare Tunnel        | ₹0           |
| API Usage                | ₹300-800     |
| ChatGPT Plus (Optional)  | ~₹2,000      |
| Claude Pro (Alternative) | ~₹2,000      |
| Optional VPS             | ₹400-900     |

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# Recommended Configurations

## Budget (₹0–500/month)

- Ollama
- MLX
- FastAPI
- SQLite
- Qdrant
- n8n
- Cloudflare Tunnel
- Minimal API usage

**Rating:** 7.5/10

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## Best Value (₹2,000–₹3,000/month)

- ChatGPT Plus
- Ollama
- FastAPI
- SQLite
- Qdrant
- n8n
- Cloudflare Tunnel
- MCP Servers
- Local image analysis
- Firecrawl
- Automatic model routing

**Rating:** 9.8/10

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## Power User (₹5,000+/month)

- ChatGPT Plus or Pro
- Claude Pro
- Gemini Advanced
- VPS
- Dedicated GPU Cloud
- API credits

**Rating:** 10/10

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# Final Recommendation

For my hardware (MacBook M1 Pro, 16GB RAM) and target budget (₹2,000–₹3,000/month), the ideal architecture is a **hybrid Personal AI Operating System**.

The MacBook serves as the always-on orchestration node, hosting local models, vector databases, automation workflows, MCP servers, and personal memory. Routine tasks—such as summarization, embeddings, image tagging, playlist management, and lightweight coding—run locally using Ollama or MLX.

When advanced reasoning or complex coding is required, requests are automatically routed to frontier cloud models (e.g., ChatGPT or Claude). This hybrid approach minimizes cost while maintaining high-quality results.

All personal context remains on the Mac whenever possible. Phone access is provided securely through Cloudflare Tunnel, and MCP enables seamless integration with Apple Music, Google Photos, GitHub, Gmail, Calendar, and future services.

This architecture offers:

- Unlimited local AI usage
- Privacy-first design
- Automatic model routing
- Secure remote access
- Extensible automation
- Excellent coding capabilities
- Intelligent photo and music management
- A scalable foundation for building a long-term personal AI ecosystem